

Conversions to memorize

Based on your four-page source PDF, with added explanations, memory aids, and practice. Learn the reference facts, study the examples, then try pages 5-6 without looking back.

kg > g > mg > mcg | Each step right: x 1,000. Each step left: divide by 1,000.

Metric: exact relationships

1 kg = 1,000 g | kilograms to grams

1 g = 1,000 mg | grams to milligrams

1 mg = 1,000 mcg | milligrams to micrograms

1 L = 1,000 mL | liters to milliliters

Household and weight: study conventions

1 tsp = 5 mL | 1 tbsp = 15 mL = 3 tsp

1 fl oz is about 30 mL | 1 cup is about 240 mL

1 kg is about 2.2 lb

Use these rounded household and weight factors for all exercises in this guide. They are the conventions in your source PDF; they are not all exact physical equivalents.

Know what the unit measures

Mass: kg, g, mg, mcg. Volume: L, mL, tsp, tbsp, fl oz. A fluid ounce measures volume. Do not treat it as an ounce of weight.

You cannot convert mg directly to mL without a concentration, such as 125 mg per 5 mL.

Smaller unit = bigger number. Larger unit = smaller number. The amount itself stays the same.

Study examples only. Follow your program, exam, or workplace conversion and rounding rules. For liquid medicines, use an appropriate calibrated dosing device, not a household spoon.

Follow the units

1. Grams to milligrams

$$2 \text{ g} \times 1,000 = 2,000 \text{ mg}$$

Check: mg is smaller than g, so the number increases.

2. Milligrams to grams

$$500 \text{ mg} / 1,000 = 0.5 \text{ g}$$

Check: g is larger than mg, so the number decreases.

3. Micrograms to milligrams

$$250 \text{ mcg} / 1,000 = 0.25 \text{ mg}$$

Check: 1 mg contains 1,000 mcg; 250 mcg is one-quarter of 1 mg.

4. Two steps: grams to micrograms

$$0.004 \text{ g} \times 1,000 = 4 \text{ mg}; \text{ then } 4 \text{ mg} \times 1,000 = 4,000 \text{ mcg}$$

Check: Two steps multiply by 1,000,000 in total, not 2,000.

5. Liters and milliliters

$$1.5 \text{ L} \times 1,000 = 1,500 \text{ mL}; 750 \text{ mL} / 1,000 = 0.75 \text{ L}$$

Check: A volume below 1 L must be below 1,000 mL.

6. Household volume

$$2 \text{ tbsp} \times 15 \text{ mL/tbsp} = 30 \text{ mL}$$

Check: 2 tablespoons = 6 teaspoons; $6 \times 5 \text{ mL} = 30 \text{ mL}$.

7. Pounds and kilograms

$$154 \text{ lb} / 2.2 = 70 \text{ kg}; 70 \text{ kg} \times 2.2 = 154 \text{ lb}$$

Check: For the same weight, the pounds number is larger.

Unit-canceling method: $2 \text{ g} \times (1,000 \text{ mg} / 1 \text{ g}) = 2,000 \text{ mg}$. Put the starting unit on the bottom so it cancels.

Desired / Have x Quantity

$D / H \times Q = \text{amount to give}$. Match the units of D and H before calculating.

D = desired dose (ordered). H = available dose (label strength). Q = the number of tablets or the volume containing H.

1. Tablets

Ordered: 500 mg. Available: 250 mg per 1 tablet.

$(500 / 250) \times 1 = 2$ tablets.

Check: 2 tablets $\times$ 250 mg/tablet = 500 mg.

2. Liquid medication

Ordered: 250 mg. Available: 125 mg per 5 mL.

$(250 / 125) \times 5 = 10$ mL.

Check: The desired dose is twice the available dose, so use twice 5 mL.

3. Convert before calculating

Ordered: 0.5 g. Available: 250 mg per tablet.

$0.5 \text{ g} \times 1,000 = 500 \text{ mg}$. Then $(500 / 250) \times 1 = 2$ tablets.

Check: Dividing 0.5 by 250 before converting would give the wrong answer.

4. A dose smaller than the label amount

Ordered: 75 mg. Available: 150 mg per 5 mL.

$(75 / 150) \times 5 = 2.5$ mL.

Check: Half of 150 mg needs half of 5 mL.

A second way: find the concentration

$125 \text{ mg} / 5 \text{ mL} = 25 \text{ mg/mL}$. For a 250 mg dose: $250 \text{ mg} / (25 \text{ mg/mL}) = 10 \text{ mL}$. This agrees with $D / H \times Q$.

Do not automatically set Q to 1. If the label says 125 mg per 5 mL, then H = 125 mg and Q = 5 mL.

Remember, calculate, check

Five memory cues

"Kind Giants Make Muffins" = Kilo, Gram, Milli, Micro. Use it to recall $\text{kg} > \text{g} > \text{mg} > \text{mcg}$. Each neighboring step is a factor of 1,000.

"Small units, big count." Imagine cutting one large piece into 1,000 small pieces: the count grows.

"Tea is 5; table is triple." $1 \text{ tsp} = 5 \text{ mL}$; $1 \text{ tbsp} = 3 \text{ tsp} = 15 \text{ mL}$.

"Pounds are plentiful." The pounds number is larger: kg to lb multiplies by 2.2; lb to kg divides by 2.2.

"Want over got, times the quantity." $D / H \times Q$. First match the units.

Fractions and decimals you will use

$1/2 = 0.5$ | $1/4 = 0.25$ | $3/4 = 0.75$

Example: $3/4 \text{ g} = 0.75 \text{ g} = 750 \text{ mg}$. Dividing by 1,000 moves the decimal three places left; multiplying moves it three places right.

Use the SAME checklist

S - State the starting unit and the unit you need.

A - Align units before dose math.

M - Multiply or divide, keeping units in the setup.

E - Evaluate: does the size make sense? Check in reverse.

Avoid easy errors

Write 0.5, not .5; write 5, not 5.0. Keep full precision during a calculation and round only at the end as instructed. The practice questions here do not require rounding.

Ten-minute study routine

2 minutes: write the conversions from memory.

3 minutes: explain two worked examples aloud.

4 minutes: solve six practice questions with units.

1 minute: correct mistakes and name the rule you missed.

Conversions

Use the factors on page 1. Write a setup and a labeled answer on each line. For rounded factors, your answer is approximate. Cover the answer key.

1. Convert 3 g to mg.

2. Convert 1,250 mg to g.

3. Convert 400 mcg to mg.

4. Convert 0.6 mg to mcg.

5. Convert 2.25 L to mL.

6. Convert 85 mL to L.

7. Convert 0.002 g to mcg.

8. Convert 3 tsp to mL.

9. Convert 2 tbsp to mL.

10. Convert 90 mL to fl oz.

11. Convert 0.5 cup to mL.

12. Convert 176 lb to kg.

Apply the math

Use D / H x Q when appropriate. Convert first if needed. All medication scenarios are hypothetical arithmetic exercises.

13. Convert 45 kg to lb.

14. Ordered: 750 mg. Available: 250 mg per tablet. How many tablets?

15. Ordered: 300 mg. Available: 150 mg per 5 mL. How many mL?

16. Ordered: 0.75 g. Available: 250 mg per tablet. How many tablets?

17. Ordered: 125 mg. Available: 250 mg per 5 mL. How many mL?

18. Ordered: 0.2 mg. Available: 100 mcg per tablet. How many tablets?

19. Ordered: 450 mg. Available: 150 mg per 5 mL. How many mL?

20. A liquid contains 100 mg per 5 mL. How many mg are in 10 mL?

21. A liquid contains 200 mg per 5 mL. What is its concentration in mg/mL?

22. Find the error: "0.5 g = 0.0005 mg." Give the correct value.

23. Find the error: for 250 mg from 125 mg per 5 mL, a student answers 2 mL.

24. True or false: 500 mg always equals 5 mL. Explain.

Check your setup

Answers using the study factors on page 1. Questions 10-13 use rounded conversion factors; those converted values are approximate.

1. $3 \times 1,000 = 3,000$ mg.
2. $1,250 / 1,000 = 1.25$ g.
3. $400 / 1,000 = 0.4$ mg.
4. $0.6 \times 1,000 = 600$ mcg.
5. $2.25 \times 1,000 = 2,250$ mL.
6. $85 / 1,000 = 0.085$ L.
7. $0.002 \times 1,000 \times 1,000 = 2,000$ mcg.
8. $3 \times 5 = 15$ mL.
9. $2 \times 15 = 30$ mL.
10. $90 / 30 = 3$ fl oz.
11. $0.5 \times 240 = 120$ mL.
12. $176 / 2.2 = 80$ kg.
13. $45 \times 2.2 = 99$ lb.
14. $(750 / 250) \times 1 = 3$ tablets.
15. $(300 / 150) \times 5 = 10$ mL.
16. 0.75 g = 750 mg; $(750 / 250) \times 1 = 3$ tablets.
17. $(125 / 250) \times 5 = 2.5$ mL.
18. 0.2 mg = 200 mcg; $(200 / 100) \times 1 = 2$ tablets.
19. $(450 / 150) \times 5 = 15$ mL.
20. $(100 \text{ mg} / 5 \text{ mL}) \times 10 \text{ mL} = 200$ mg.
21. $200 \text{ mg} / 5 \text{ mL} = 40$ mg/mL.
22. Wrong direction: g to mg multiplies. $0.5 \times 1,000 = 500$ mg.
23. The student omitted Q = 5 mL. $(250 / 125) \times 5 = 10$ mL.
24. False. You need the concentration to convert a mass to a volume.

Sources: User-provided pharmacy_tech_math_conversions_study_guide.pdf, pp. 1-4. Added mnemonics, examples, and exercises follow its scope.

Measurement and decimal notation: FDA, Dosage Delivery Devices for Orally Ingested OTC Liquid Drug Products (2011); NCPDP metric dosing resource hosted by FDA.

[FDA guidance](#) | [Metric dosing resource](#)